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Your AI Produces Outputs. Which Ones Count?

FinOps, Tokenomics, and Accepted Work per Dollar

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Kenneth L. Cheney · Expound Laboratories™ · August 2026

1. Barcelona

In October 2018, Jay Wampold, J.R. Storment, Adam Heher and I spent a week in a hotel above the Mediterranean in Barcelona, working out how to assess the economics of a cloud estate. Four of us, three whiteboards, and a city that made it easy to keep going.

Arguing was the method. Someone would put a question on the board, someone else would explain why it was the wrong question, and we would take it apart over paella and cava and put it back together differently. The boards accumulated in layers, red in the morning and blue over it by evening. Very little survived in the form it arrived in.

What we ended up with were headings. **Allocation. Rates and reserved instances. Optimization: usage, rate, elasticity.** And in the corner of two boards, in green, the rubric we scored everything against: *bad, good, question, what good looks like, differentiator, example*. We did not think we had invented a category. We thought we had something worth testing with other people.

The problem had become impossible to ignore. The people who caused technology spending were no longer the people who could see it, approve it, or explain it. An engineer with an API key could commit real money in an afternoon. Finance found out at the end of the month, in a bill with the granularity of a phone book and the interpretability of a weather map. Technology accounting assumed expenditure was a decision you made occasionally and lived with for years. Cloud made it one you made continuously and could reverse in minutes.

What we kept circling was that this was not an accounting problem but an operating one. The organization needed a shared language in which an engineer, a finance analyst and a product owner could look at one number and mean the same thing by it. Without it, every conversation was the same argument: engineering said the spend was necessary, finance said it was unmanaged, and neither could resolve it because they were not describing the same thing.

Then we did what I had done for years with every framework I worked on. We tested it against everyone who would have to live with it. Customers. Partners. Analysts who had to explain the market to buyers. Influencers who would either repeat it accurately or not at all. Colleagues inside Cloudability who would have to sell it, support it and defend it. We asked what they counted, what they argued about, what they could not defend in a review, and where the numbers stopped being useful. Then we iterated again.

That broader work became the initial FinOps framework and the taxonomy. I was also part of the larger effort that created the category and launched the Foundation, alongside others.

The four names above identify the people who were in the room in Barcelona. They are not a list of the people who built what followed.

The framework and the taxonomy came out of that work. The discipline did not. The discipline is what a great many people have built on the framework since, most of them on problems we never thought of, and it is much larger now than what we handed them. The Foundation's own account of its founding starts in Cloudability's Customer Advisory Board meetings, where practitioners kept saying they needed a community, and its founding members were operators at companies with nothing to sell. If you have carried this work since 2019, the part that matters is yours, not mine.

I raise it for one reason, and it is not credit.

Every question we wrote down that week governs the same kind of thing. How to allocate shared costs. How much of an estate reserved instances cover. Whether to rightsize across instance families or only within them. What to do about untaggable spend. By the last afternoon we had reached toward business value too, mapping cloud spending onto the taxonomy of the business, the instinct that later became unit economics. Good questions. I would still ask most of them today.

None of them defines what would admit one particular output into a productive numerator.

That was not a defect in the problem we were solving. Quality, review, audit and professional accountability sat in adjacent disciplines rather than inside the cost framework, and there was no reason to pull them in. But the omission is not somebody else's oversight to correct. It was ours, it was reasonable at the time, and AI has now made it economically load-bearing.

In the cloud era, the bill was the imperfect proxy. In the AI era, the bill is still an imperfect proxy, and so is the output.

2. The second variable-cost revolution

Consider what an organization now buys with AI capability. Model inference, priced per token at rates that differ by an order of magnitude across tiers and change without notice. Retrieval (embedding, indexing, vector search, reranking), with its own infrastructure and per-query cost. Tool invocations, some internal, some metered third-party APIs. Orchestration, the harness that decomposes a request, fans it out, retries failures and decides when to stop. And context, which is not free and grows as trajectories lengthen.

It also buys what appears on no AI invoice: retries, because the first attempt was wrong in a way the system could detect; verification, because it was wrong in a way nothing downstream could detect; and human review, correction, escalation, reconciliation and audit reconstruction six months later.

The economically consequential change is the relationship between production and checking cost. In most prior automation, producing a candidate answer was expensive relative to checking it, which is why review was affordable. Generative systems inverted that: producing a plausible candidate became very cheap; establishing whether it can be relied upon did not.

Production scaled. Assurance did not.

That divergence between production and assurance shows up first as a review queue. Then as a quiet decision to stop reviewing, usually justified by a sampling argument that made sense at lower volumes. Eventually as an incident, a restatement, a regulatory finding, or a customer who acted on something the organization never stood behind.

Agentic systems sharpen the divergence. When a model produces text, a human sits between output and consequence. When an agent invokes a tool, that interval closes: the refund is issued before anyone has decided whether the work qualified.

AI is therefore the urgent case. Not because AI is the only place the problem exists (it is not, and section 11 makes that argument), but because it is where the gap between what a system produces and what an organization can stand behind has widened fastest.

3. Start with the strongest version of what already exists

It would be easy, and wrong, to open by claiming the industry is naively counting tokens. Whatever was true when the first per-token invoices arrived is not true now, and a case built on it would be a strawman.

FinOps has already reached outcome economics. The Foundation's current definition describes a practice that "maximizes the business value of technology," and the site is blunt about the misconception: "If it seems that FinOps is about saving money, think again." The Unit Economics capability already distinguishes resource efficiency metrics, "cost per GB stored... cost per token," from business unit metrics, "cost per tenant, cost per transaction, cost to serve, cost per case resolved." Its maturity model runs from technical unit costs at Crawl to a Run stage where "unit metrics incorporate fully loaded costs across all applicable technology categories."

Cost per case resolved is already in the FinOps framework. Fully loaded costs are already the mature target. AWpD therefore cannot credibly claim to introduce outcome-oriented unit economics to this audience. Its contribution lies elsewhere.

Tokenomics has already reached full AI cost. The Tokenomics Foundation defines its subject as "the discipline of converting energy and capital into AI, then efficiently consuming AI services to enable intelligent outcomes and drive business value," and states plainly: "Tokenomics isn't about counting tokens. Tokenomics connects the total cost of AI (including labor) to its outputs."

Including labor. To outputs. That is a serious economic frame, built by a vendor-neutral body: the Foundation launched on August 4, 2026, hosted by the Linux Foundation, with thirty initial members. Its published work includes Big-T notation for how token consumption scales. Its named workstreams include a reference model for the full cost of AI, a cost-to-serve methodology, a value measurement framework, token cost telemetry aligned to a FOCUS release due December 2026, and an **AI Unit Economics Measurement Framework**. That is the nearest work to anything here, and the Foundation itself is three weeks old at the time of writing. AWpD should consume that work, not compete with it.

So let me be precise about what I am *not* claiming. Cost per token is not wrong. It answers an upstream question, what a unit of model consumption cost, and answers it well. FinOps does not stop at infrastructure. Tokenomics does not stop at tokens; it says so itself. And including human, tool and infrastructure cost in a denominator is not a novel contribution of mine. Both disciplines already do it.

What I am claiming is narrower. Both disciplines are building increasingly good answers to *what did it cost*. What I did not find in the public materials I reviewed in August 2026 is a governed per-work-item **acceptance** object: a record, for a particular output, that an authority independent of the producer admitted it, and a corresponding cost line for the work of deciding.

There is a nearby thing that does exist. FinOps does ask for governance; the Unit Economics capability names "clear ownership and governance of definitions" at its Run maturity, a real requirement seriously stated. But governing the definition is governing *what you have chosen to*

count, not an authority, separate from the party being measured, deciding item by item whether a given output qualified to be counted at all. The first makes your metric consistent. The second makes it hard to fake.

That question sits directly between cost and value:

What authorizes an output to enter the productive numerator at all?

Every unit-economic measure has a numerator that counts something. Cost measurement has become materially more sophisticated: more complete, more attributable, more fully loaded. In many operating metrics, however, the rule admitting something *into* the numerator is still inherited from a completion status, a workflow disposition, or an evaluation definition. Outputs. Completions. Closed tickets. Green statuses.

That was a reasonable inheritance when production was slow enough that a human looked at nearly everything. It is not reasonable now.

You can test this without adopting anything in this paper. Take one unit metric your organization already reports.

1. **What admitted each item to its numerator?**
2. **Who or what established that the item qualified, and were they independent of its producer where the consequence required it?**
3. **Does that permission still apply, to the party relying on it and for the purpose they are relying on it for, now?**
4. **Did the denominator keep every attributable path to the disposition, including failures, refusals, retries, review and remedy?**

Then ask: **if those answers are unclear, what exactly is the reported productivity metric measuring?**

4. Follow one dollar downstream

The chain to hold in view:

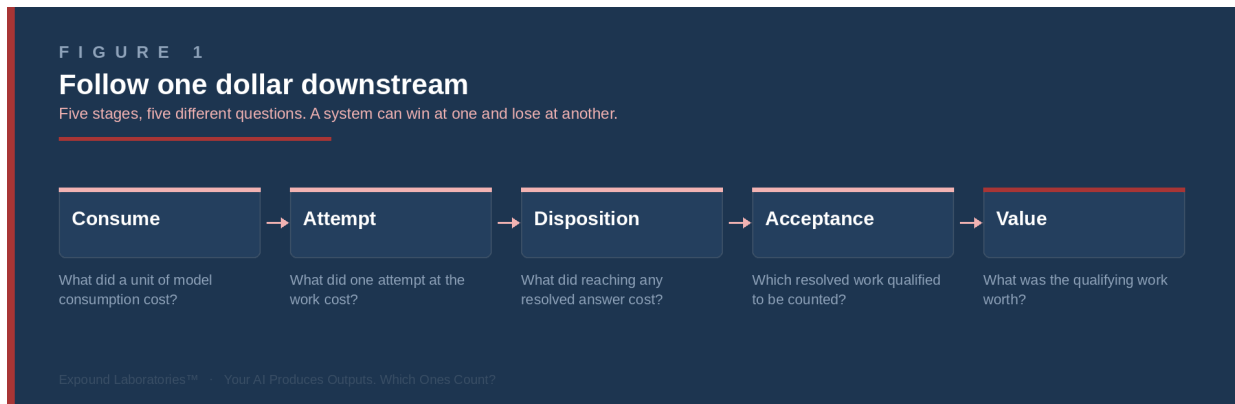


Figure 1 Follow one dollar downstream

Five stages, five different questions, and **a system can win at one stage and lose at another.**

The case I carry through the paper:

An enterprise runs a customer-support operation. A class of cases requires determining whether a customer is entitled to a billing adjustment, and if so how much, from contract terms, transaction history, applicable policy version, and prior adjustments. Get it wrong one way and you have given away money you did not owe; the other way, a complaint, an escalation and possibly a regulator.

Two vendors propose AI-assisted architectures.

Architecture A uses a large frontier model with a long context window. It ingests the account history, reasons over it, and produces a determination with an explanation. It is expensive per call, and all else equal resolves a wider range of cases in a single pass than a smaller model would.

Architecture B uses a small, cheap model as a classifier, routes to deterministic retrieval for the contract terms and transaction history, computes the adjustment arithmetically, and uses the model only to draft the customer-facing explanation. It is dramatically cheaper per call, gets a narrower set of cases right, and declines or escalates more often.

Consume. B wins overwhelmingly: perhaps a twentieth of A's token consumption.

Attempt. Still B, but by less. A resolves more cases in one attempt; B retries and re-routes more.

Disposition. A produces confident determinations that sometimes cite policy language absent from the version applicable to that account. Catching that requires a verification step routing the case to a human. B's declines are cheap and immediate, and its accepted cases carry deterministic arithmetic needing no re-checking. Depending on how often A's plausible-but-wrong determinations occur, the cost of reaching a *resolved* disposition can invert the ranking.

None of which is an argument against capable models. Premium models earn their incremental cost where they materially improve the rate at which work reaches an accepted disposition, and on harder case mixes they frequently do.

Acceptance. Of the cases each resolved, how many met the conditions the organization requires before acting? Correct arithmetic. Applicable policy version. Adequate transaction evidence. Authority appropriate to the adjustment size. A record sufficient to explain the decision later.

Value. Which architecture produced more satisfied customers, fewer complaints, better retention, lower regulatory risk? Not answerable from any of the above.

The failure mode is using an upstream measure to answer a downstream question. Each hop is a small, reasonable-looking extrapolation, and the errors compound.

So which should the enterprise buy? The question is underspecified, and that is the point. Under an acceptance-qualified numerator B may win again, not for capability but for evidence. The question that can reverse the decision is the one nobody priced. In most procurement conversations I have seen, only the first of the five stage questions gets asked.

This paper is about the two stages with the least measurement discipline: **Disposition** and **Acceptance**.

5. Cost-to-Disposition

Take the denominator first. FinOps practitioners have most of the machinery, and the idea is about fifty years old.

The Cost of Quality tradition, formalized through the American Society for Quality, divides the cost of getting work right into four categories: **prevention** (so defects do not occur), **appraisal** (inspection, verification and audit to find out whether they did), **internal failure** (what defects cost before reaching a customer) and **external failure** (what they cost after). Regulated industries have costed work this way for decades, and the boundary list below maps onto them: verification is appraisal, retries and rework are internal failure, remediation is external failure.

Cost-to-Disposition composes those antecedents around a different measurement object. Cost of Quality does not itself constitute the eligible population, fix the identity of one attempted item, treat a correct refusal as a disposition rather than a defect, account for abandoned or indeterminate work, establish acceptance authority, or bound what the number entitles anyone to claim. It supplies the insight that verification and failure cost money; this paper applies that to one constituted unit of work and every attributable path to its disposition.

Cost-to-Disposition is everything economically attributable to reaching a resolved disposition about a unit of work, whatever that disposition turns out to be.

Not the cost of success. The cost of *resolution*.

For an AI-assisted work class, the boundary typically includes:

- model inference across every attempt, not only the successful one
- retrieval, embeddings, vector storage and reranking
- tool and API invocations, including metered third-party calls
- orchestration and harness overhead
- retries, system- or checker-triggered
- verification, including second-pass checks
- human review time, at loaded cost
- escalation handling
- correction and rework
- reconciliation where systems disagree
- audit reconstruction where evidence was not captured
- remediation where something reached a customer

Three rules make this tractable.

Failure does not erase expenditure. A declined case, a failed attempt and an abandoned trajectory all cost money. They belong in the denominator of the work class that produced them.

The final successful attempt does not make the earlier attempts free. Systems that log only the successful call and discard the failures can materially understate the cost of producing the disposition.

Cost follows causal contribution, not the organizational chart. Verification that moved to another team did not disappear, and correction that moved to the operations budget was still incurred by the work class. A measure that improves because cost moved to a budget you do not read is not an improvement.

Two boundaries must be declared rather than assumed. **Marginal versus fully loaded:** inference-only cost answers what one more costs now; fully loaded cost answers what the capability costs to operate; and comparing across them is meaningless. **Horizon:** a disposition reached today may generate remediation cost next quarter, so three-day and ninety-day boundaries rank architectures differently, and the shorter one flatters systems whose failures surface late.

Cost-to-Disposition is not "fully loaded cost" renamed. Fully loaded cost is a question of **breadth**: which categories are included. That is ground FinOps and Tokenomics already push hard on. Cost-to-Disposition is a question of **integrity**, whether the accounting can be rearranged without anyone noticing.

The nearest neighbor is cost to serve. Tokenomics is directly addressing AI cost breadth and cost to serve, and this paper assumes that work rather than competing with it. Cost-to-Disposition asks a different question: what did the complete work trajectory cost before the organization reached any disposition, including paths that failed, refused, escalated or remained unresolved? Cost stays bound to the eligible work and every path to its disposition.

Two organizations can include the same categories and still produce incomparable numbers, differing on attempt identity, eligibility, retry attribution, verification visibility, horizon and comparison basis. **That is why the denominator remains load-bearing in this framework rather than being a generic TCO claim.** Cost breadth is increasingly mature. Disposition integrity is a separate problem.

6. The hardest word is "Accepted"

Anyone can divide outputs by dollars. The interesting question is what belongs in that numerator, and most organizations have never decided. They inherited it.

Accepted Work is work admitted to count because predetermined conditions were affirmatively established for a stated purpose.

Six non-equivalences do most of the teaching:

Generated is not accepted. The system produced something. That is a fact about the system, not about the work.

Correct is not accepted. A billing determination can be arithmetically perfect and still fail to qualify, because the organization requires evidence of which policy version applied and the system did not retain it. Correctness is a property of the answer; acceptability is a property of the answer *plus* what can be established about it.

Completed is not accepted. A status field is a claim someone wrote, usually the producer, usually without independent verification. Workflow completion means the workflow advanced.

Producer-approved is not independently acceptable. When the same system, team or model both produces the work and grades it, the grade is a producer report. It may be entirely honest and still be the wrong evidence class for a consequential decision. Regulated supply chains accept

a supplier's own certificate of conformance, but only inside a qualification and surveillance regime that establishes what the certificate is worth. In software we routinely accept the certificate and skip the regime.

Accepted once is not accepted now. Acceptance rests on conditions: a policy version, an authority, a dependency, an evidence artifact. Conditions change. Work accepted in March under a policy superseded in June may no longer qualify, and in most systems nothing recomputes.

This is the point I would most like a reader to take away, because it is the one that separates a governed numerator from a historical one. Work is not permanently admitted merely because it once passed. Where the applicable evidence, authority, policy, dependency or relying purpose changes, current acceptance may narrow or be withdrawn. The historical fact of acceptance stays in the record, and should. It does not silently remain present authority.

Accepted is not valuable. Qualifying work can be worth nothing. Acceptance is a gate, not a valuation.

Which raises the right question:

Who decides that the work counts?

For low-consequence work, the honest answer is: whoever normally decides. If nobody will rely on it and nothing follows from it, ordinary acceptance is fine and additional machinery is waste.

Governing everything fails as surely as governing nothing.

For consequential work, the answer has to be better: the producer's assertion that it finished is not the same class of thing as an established basis for reliance. This is where the Finality Assurance Family of Standards (FAS) enters, kept to what it is:

FAS computes a present, scoped reliance decision (what a named party may rely upon, about a named result, for a named purpose, under current conditions) rather than permitting the producer to write that decision directly.

Three properties of that formulation do the work. It is **present-tense**: a decision about now, recomputable when conditions change, not a permanent stamp. It is **scoped** to a party and a purpose, because reliance is not a global property. And it is **computed rather than asserted**: derived from evidence under a predetermined basis, not written into a field by the thing being evaluated.

The public FAS architecture adds one point that matters for this paper's reach: **the need for a reliance decision follows the consequence, not the identity of the executor.** A model, an agent, a rules engine, a batch job, an industrial controller and a person all produce work that something

downstream will act on, and where the consequence is serious the assurance question is the same. That is the hinge that takes accepted-work economics beyond AI.

I am deliberately not describing how any of this is implemented. This paper contains no algorithms, no scoring, no calibration, no schemas, no security architecture and no anti-bypass mechanics. Those are reserved. The economic argument does not require them, and a reader should be able to evaluate it without them.

7. Accepted Work per Dollar

Accepted Work per Dollar is the quantity of qualifying work produced per unit of complete attributable expenditure.

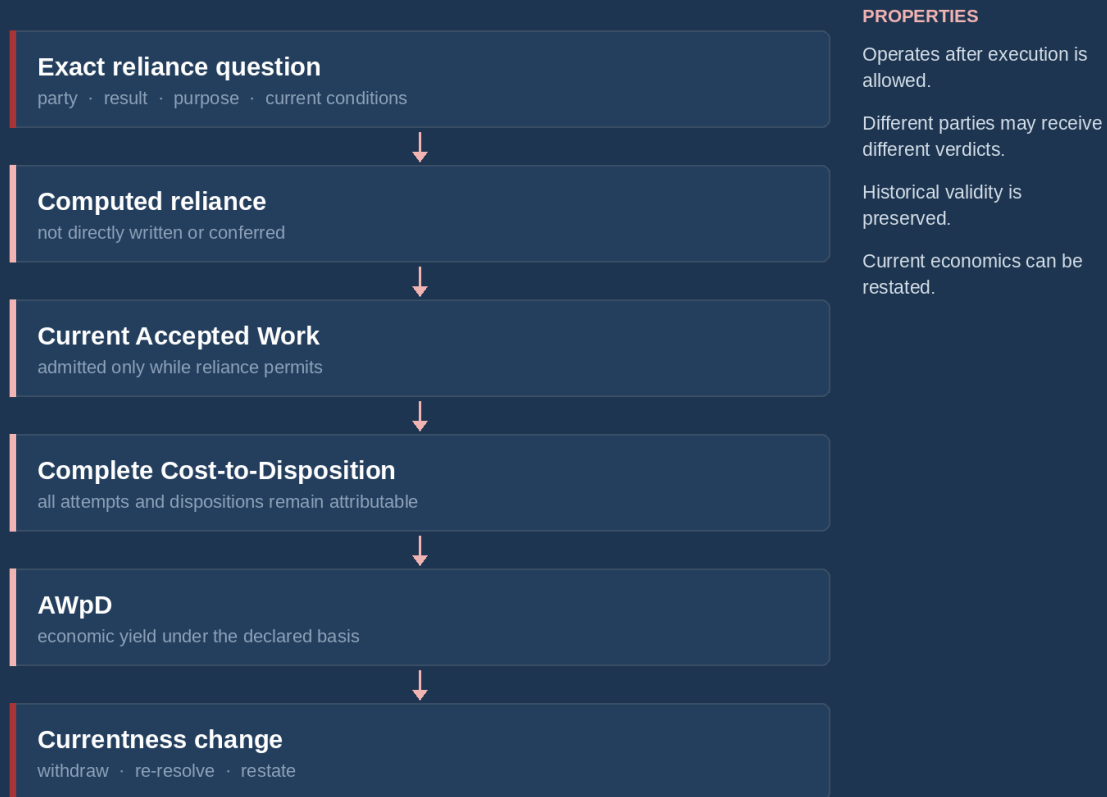
Numerator: work admitted under a predetermined acceptance basis. Denominator: Cost-to-Disposition.

One separation matters most, because collapsing it is the most likely way to misread this paper. **These are two different objects, and one governs the other.**

FIGURE 2

The complete FAS-to-AWpD loop

The distinctive object is the complete loop, not any one component.



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Figure 2 The complete FAS-to-AWpD loop

FAS computes reliance. AWpD computes economics. The separation is not tidiness. It stops economic pressure, the measured workflow, or an optimizer from defining its own numerator.

No reliance computation, no governed Accepted Work. No governed Accepted Work, no conformant AWpD.

That is a statement about the logic, not a requirement to buy anything before starting. An organization can begin today by constituting its work, declaring the population, writing the acceptance basis down, separating the authority that applies it, measuring Cost-to-Disposition and computing the ratio. What that alone does not produce is computed reliance, a present, party- and purpose-scoped decision about a named result. That takes a mechanism built to compute it, and the mechanism is not hypothetical. Section 17 names what is available and what is planned.

AWpD has three identities, and confusing them causes most of the misuse I have seen.

As a concept, it asks a question every executive already has: *how much of what we bought counted?* No special vocabulary is needed to feel its force, which is why it opens well with a CFO.

As a measurement discipline, it is a set of things that must be fixed and evidenced before the number means anything: the bulk of the work, and the subject of sections 8 and 9.

As an operating signal, it informs allocation among *admissible* alternatives: routes, architectures, model tiers, degrees of automation, human involvement. Note the qualifier. AWpD is a tiebreaker among options that already satisfy the organization's constraints. It is not an argument for choosing an inadmissible option because it is cheaper per accepted unit.

The most important sentence here is a warning:

A number labeled AWpD without a governed work unit, a defined eligible population, a predetermined acceptance basis, a declared cost boundary and an established comparison basis is just another productivity ratio. And a productivity ratio becomes easy to game when the measured party can change the work unit, the population, the acceptance rule, the cost boundary, the horizon or the comparator without changing the identity of the reported metric.

AWpD is a measurement discipline with a substantial development record behind it, and it is implemented: the acceptance architecture computes reliance, and the economics compute from it. Section 15 states what the measure does and does not establish, and a reader should meet those boundaries early rather than discover them late.

8. How a true metric becomes a false economic story

Before asking what protects AWpD from gaming, there is a prior question the industry mostly skips:

Why is it so easy to improve the metrics we use today without improving the economics those metrics are being used to represent?

The answer is not that practitioners are dishonest, or that the metrics are defective. Most were designed to answer narrower upstream questions, and become easy to game *as measures of productive work* when boundaries are left open.

Three different things get called "gaming," and conflating them is how this conversation goes wrong.

Legitimate local optimization. Cost per token falls because the team improved caching, trimmed context, or moved suitable work to a smaller model. Nothing has been gamed: an upstream measure improved within its proper office.

Proxy optimization. Cost per token falls; retries rise; human correction rises; slightly less work satisfies acceptance. The token metric is *still accurate*, but it has never been a measure of accepted-work productivity.

Claim laundering. "Our token cost fell 40 percent, therefore our AI productivity improved 40 percent." The first clause may be entirely true; the second does not follow.

The metric is often true. The economic story told with it is not.

Where the gaming surface is

FIGURE 3

Where the gaming surface is

Common productivity metrics, what they legitimately answer, and which lock closes the manipulation.

METRIC	LEGITIMATELY ANSWERS	BENIGN IMPROVEMENT THAT STAYS VALID	NUMBER IMPROVES WITHOUT MORE ACCEPTED WORK	BASIS CHANGED, LABEL KEPT	CANNOT SEE	LOCK
Cost per token	What did one unit of model consumption cost?	Better caching, trimmed context, right-sized model	Cheaper model while retries, review or failure rise	Changing which tokens count (cached, system, failed, retried) or marginal vs loaded rate	Complete workflow cost; accepted yield	Cost + Acceptance
Tokens per task	How token-intensive was the declared task?	Genuinely fewer tokens for the same task	Moving work to tools or humans	Redefining the task boundary; omitting subagents	Total effort across the route	Work + Cost
Cost per request	What did one invocation cost?	Cheaper invocation, same trajectory	Splitting or combining calls differently	Omitting retries and orchestration from the request	The attempt trajectory and its disposition	Work + Cost
Cost per case resolved	What did a declared completion cost?	Same population and basis, lower cost	Dropping difficult work	Narrowing "resolved"; excluding reopened cases	Acceptance integrity; coverage	Acceptance + Population + Cost (horizon)
Tasks completed / tickets closed	How much recorded activity finished?	More work genuinely finished	Fragmenting work; prioritizing easy cases	Closing early, a status the producer writes	Whether the meaningful unit completed	Work + Population + Authority
Automation / touchless rate	How often did the process avoid an identified human touch?	Real automation with no displaced cost	Moving review downstream	Hiding exceptions; lowering escalation thresholds	Human reconciliation; false acceptance; real coverage	Cost + Acceptance + Population
Accuracy / benchmark score	How often did outputs satisfy a benchmark definition?	Genuinely better outputs on a fixed benchmark	Selecting favorable work	Averaging across different cohorts; optimizing to the benchmark	Reliance on an individual result; cost; coverage	Population + Comparison, and see the gap below
First-pass yield	What passed without identified rework?	Fewer real defects, unchanged process	Moving rework outside the observed stage	Redefining "pass"	Later failure; currentness	Acceptance + Cost (horizon)
Productive output per dollar	How much output meeting the stated definition did spend produce?	More qualifying output, unchanged definition	Threshold changed after results	Producer defines productivity	Numerator authority; complete cost	Authority + Acceptance + Cost
Utilization	How fully was a resource used?	Higher utilization on work that needed it	Premium resources busy on work that did not need them	Redefining the denominator of "available"	Accepted output; opportunity cost	Acceptance + Cost

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Figure 3 Where the gaming surface is

One boundary the locks deliberately do not cover. Optimizing to a benchmark (training or tuning toward the target rather than the underlying capability) is a property of how a system was built,

not of how its work was counted, and no measurement boundary settles it. Population and Comparison constrain *which* work is measured and *whether* two results may be compared. Neither is intended to certify that the system under measurement was not shaped to the measure. That is the job of benchmark integrity, a separate control layer, and section 9 states what it requires.

The "cannot see" column is the point of the table, and its contents are my argument rather than an observation of fact. **These metrics are useful inside their jurisdiction. The gaming surface appears when they are promoted into a claim about accepted productivity without governing the boundaries that claim requires.**

Four outcomes for the same work class

Return to the billing-adjustment case from section 4, and hold the acceptance basis fixed.

Case A: genuine improvement. Token cost falls. Retries do not rise. Human review does not rise. The same or more work satisfies the same acceptance basis. Complete Cost-to-Disposition falls.

→ *Tokenomics improves and AWpD improves. This is real end-to-end progress, and both measures should say so.*

Case B: upstream improvement only. Token cost falls. Accepted Work is unchanged. Complete Cost-to-Disposition is unchanged, because the savings were small relative to verification and human time.

→ *Token efficiency improved. Accepted-work productivity did not. Both statements are true, and neither contradicts the other.*

Case C: local metric improves, economics worsen. Token cost falls. Retries and verification rise. The system quietly avoids difficult cases. Less eligible work satisfies acceptance. Human correction moves to the operations budget.

→ *The token metric improved while accepted-work economics got worse. The token number is still correct.*

Case D: measurement gaming. Facing pressure to show the improvement, the team counts only the successful final attempt, removes the unattempted difficult work from the population, moves review cost outside the boundary, narrows the definition of "resolved," and reports against an easier baseline.

→ *The apparent improvement came entirely from changing the measurement around the work. Under conformant AWpD, four of those five changes surface as a different measurement identity,*

a conformance failure or a refused comparison. The fifth (reporting against an easier baseline) is discussed below, and it is the one the lock set closes least well.

Case B is the most common and the least discussed. **An upstream metric improving without AWpD improving does not make the upstream metric false, and it is not a failure of the team.**

Six freedoms, six locks

Why is this structurally easy? Because a conventional metric typically leaves one or more of the following open, not through carelessness but because it was never designed to answer the full accepted-work question:

5. **Unit freedom:** the measured party can redefine what one item means.
6. **Population freedom:** it can exclude difficult, refused, failed or unattempted work.
7. **Acceptance freedom:** it can change what success means, after seeing results.
8. **Authority freedom:** the producer can grade its own output.
9. **Cost and time freedom:** it can omit costs, move them elsewhere, or stop measuring before failures surface.
10. **Comparison freedom:** it can choose an easy baseline, a favorable work mix, or an incompatible comparator.

Change the basis, change the measurement.

AWpD is only AWpD when the locks are in place.

9. The Six Locks

What stops someone from improving AWpD without improving anything real? "It is designed to resist gaming" is not an answer.

Gaming is an improvement in the metric without a corresponding improvement in the economic object the metric represents.

Note what that excludes: producing more qualifying work per dollar is not gaming, it is the goal. The problem is moves that change the *accounting around* the work while leaving it unchanged.

The Six Locks expose and constrain the six first-order boundaries above: a teaching device, a synthesis I am introducing here, not a term of art from anywhere else.

They are necessary. They are not sufficient by themselves, and the end of this section says why. In conformant AWpD they operate inside a governed **Measurement Constitution** that versions the

basis, controls who may change it, and makes currentness an observed property rather than an assumption.

The locks define the boundaries. The constitution governs the boundaries.

Lock the work. Declare in advance what constitutes one unit. The classic attack is decomposition: split one case into four sub-cases, accept three, and report 75 percent acceptance on work that was one unit. Aggregation does the reverse, hiding failure inside a larger unit that passes on balance.

Lock the eligible population. Declare which work is in scope before you see what the system handled. A system that quietly declines difficult cases can post excellent AWpD while providing less capability than the one it replaced. That is the most economically damaging attack in the set. **High AWpD does not imply broad coverage.** Eligible, attempted, unattempted, accepted, refused and indeterminate work must reconcile.

Declared scope is not enough: composition inside it can drift without anything being excluded. If this quarter's population is easier than last quarter's, AWpD rises and every lock passes while nothing improved. So the lock requires **composition to be reported, not merely scope declared**, and reported stratified by difficulty band or whatever proxy the work class supports. A single ratio over a shifting mix is not a measurement.

Lock acceptance. Fix the conditions before outcomes are known. Criteria adjusted once the results are visible are narration, not measurement. But fixed conditions are *stable*, not necessarily *adequate*: a weak basis declared honestly in advance passes this lock while inflating the ratio. Stringency is domain-specific, so **the acceptance basis travels with the claim**, and a weak basis limits what may be claimed rather than quietly raising what is reported.

Lock acceptance authority. Establish who or what may determine that conditions were met, with separation appropriate to the consequence. Without it the rest are decorative: the producer can satisfy them all on paper. Separation must extend from *assertion* to *construction*, because if the acceptance evaluator is a model the producing team selected, prompted and tuned, the producer is still grading itself. For AI work that is the dominant case, so what matters is who **builds and configures** the acceptance mechanism, not only who signs it.

Lock the cost boundary. Declare what is in and out before measuring, and hold it constant across anything you compare: retries, verification, human time and failed attempts in or out, horizon length, marginal or loaded. Each is a legitimate choice and an illegitimate surprise.

Lock the comparison. Two valid measurements do not automatically support a comparison. Comparability is a property of the *specific claim being made* and should return an explicit verdict

rather than be assumed from a shared metric name. The lock also governs **which** comparator is chosen, since a flattering prior period or weak incumbent improves the reported delta without touching the work and no compatibility test catches it. The baseline must be **pre-declared, with a stated selection rule**.

What is borrowed and what is added

Several of the locks have recognizable antecedents. FinOps Unit Economics already requires meaningful unit definitions, fully loaded costs and governance of them; Allocation handles apportionment and cost coverage; benchmarking practice already cautions that every organization's usage is unique; FOCUS supplies a common cost-data shape; Cost of Quality makes verification and failure economically visible.

Two more deserve naming, because a practitioner will reach for them first. FinOps governance capabilities already require accountability, escalation paths and continuous monitoring of whether controls are working. That is real governance, and it governs spending decisions and policy compliance rather than whether one particular output may enter a numerator. And the Framework's AI work already recognizes the importance of separating successful model outputs from unsuccessful ones. That reaches directly toward the numerator problem, and I would rather point at it than pretend the community has not noticed. In the material I reviewed, it does not create the item-level, present-tense reliance computation or the independent acceptance authority described here.

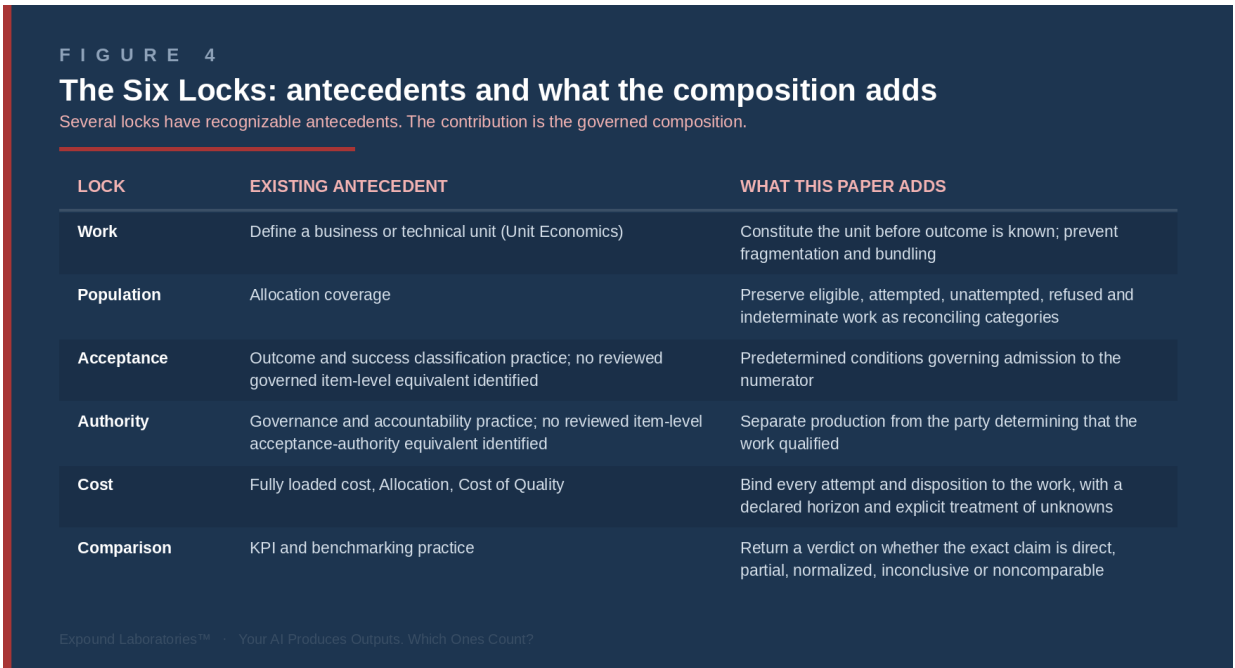


Figure 4 The Six Locks: antecedents and what the composition adds

The contribution is not six unrelated inventions. Several have recognizable antecedents, as the table shows, and the two with no reviewed item-level equivalent still sit near existing outcome-classification and governance practice. What I did not find in the reviewed material is the complete item-level reliance and acceptance object, or the composition itself. The contribution is that composition: all six travel together in a conformance-controlled Measurement Constitution and jointly determine what may enter the numerator and what the number entitles anyone to claim. Removing any one makes the rest cosmetic.

These attacks are concrete and mostly not malicious. They happen when a metric acquires consequences and no one specified the boundaries:

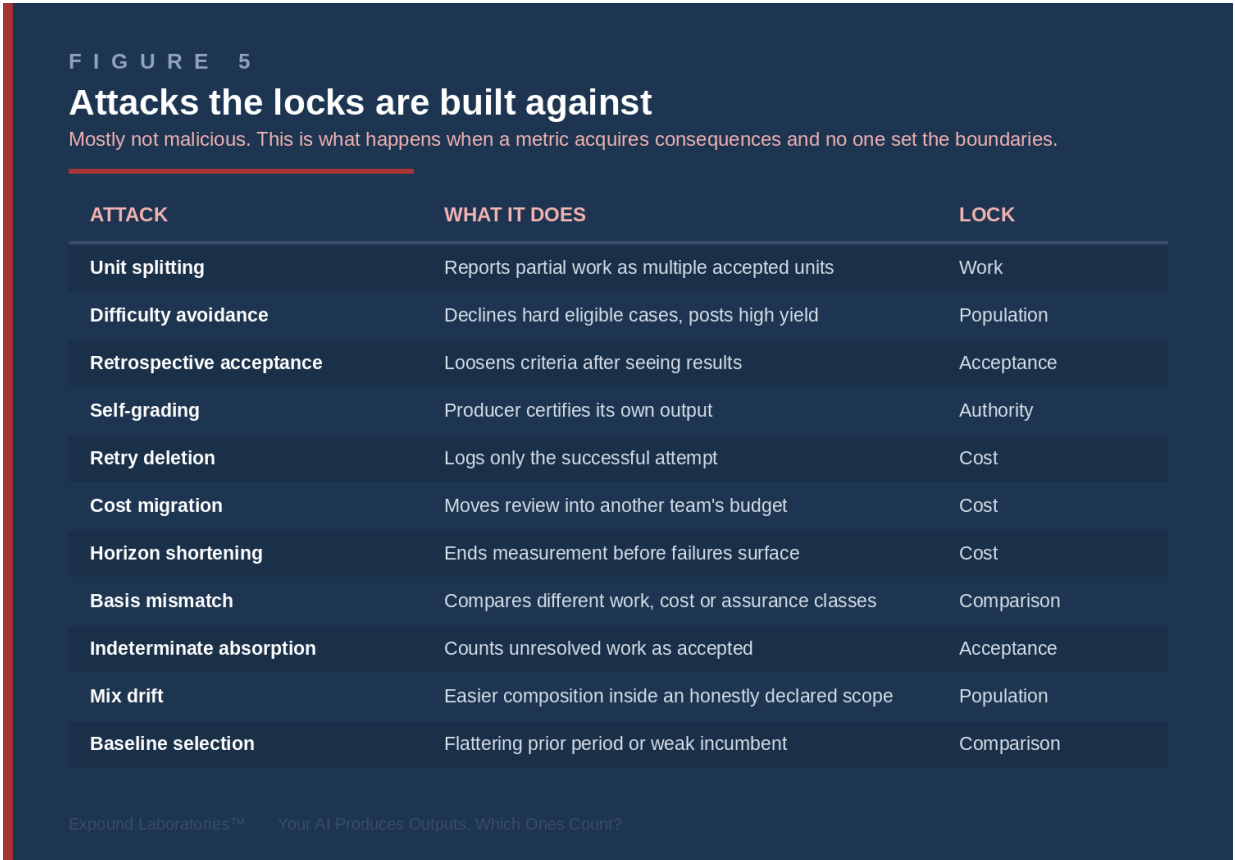


Figure 5 Attacks the locks are built against

The claim I am prepared to defend:

The Six Locks alone are necessary but insufficient. Inside the governed **Measurement Constitution** described above:

Conformant AWpD is designed so that changing the accounting around the work cannot silently improve the same licensed claim. The change must surface as a different

measurement identity, a stated limitation, a conformance failure, an indeterminate result, or a refusal to support the comparison.

Read what that does not say. It does not say AWpD cannot be gamed: any measurement can be gamed by someone willing to falsify inputs. It says the *specific* moves above should become **visible** rather than **profitable**, changing what you are permitted to claim rather than what you appear to have achieved.

One further principle:

Unknown is not zero. Unknown is not pass.

Missing cost, unresolved dispositions, unevaluated work and unavailable evidence must stay distinguishable from measured zeros and successes. The temptation in every reporting system is to coerce the unknown into whichever bucket makes the report cleaner, and that coercion destroys more measurement integrity than every deliberate manipulation above.

Where the Six-Lock abstraction stops

The Six Locks are a teaching abstraction, not the whole conformance architecture. I had the abstraction attacked as though it were sufficient by itself, from inside the rules rather than by falsifying inputs. Three attacks got through. That result matters because it shows exactly what the abstraction leaves out.

The locks identify the boundaries that must be governed. They do not, by themselves, determine who may constitute those boundaries, how a material change affects measurement identity, or what establishes that a supposedly current basis was actually observed. Those are second-order governance questions, and in the fuller architecture the locks operate inside a versioned Measurement Constitution that answers them.

The locks are mostly timing rules, and timing rules can be re-declared. Five of the six say declare in advance. None of them says where the boundary has to sit, and none limits how often you may move it. An adversary who waits a period, then prospectively re-declares a narrower unit, a shorter horizon and a re-grained population, satisfies every lock and improves the number without producing more accepted work. The cure is not another lock. **A material change to the measurement basis creates a different measurement identity. It cannot silently extend the same historical series.** New learning properly produces a new basis. What it does not do is rewrite the rules governing a result already observed. A longitudinal view across a material basis change requires explicit reconciliation. It cannot be presented as an uninterrupted trend.

You are allowed to improve the measurement. You are not allowed to improve the measurement and pretend you did not change it. Change the basis, change the identity.

Independence is required at one joint and asserted at five. The authority lock separates the party that grades the work. The producer still declares the unit, the population, the cost boundary, the horizon, the comparator selection rule, and that each was declared in advance. Those are exactly the kind of self-written status fields section 6 warns you not to trust. A measurement is only as independent as its least independent declaration. The principle is that **the stronger the claim, the stronger the required separation between the work being measured, the authority defining the measurement, the authority deciding acceptance, and the authority deciding what the evidence permits the organization to claim.** Requiring an outside party for every internal diagnostic would make the discipline unusable, so as a teaching profile the burden scales with the claim:



Figure 6 Claim strength cannot exceed constitution independence

Currentness is not a fourth class but a property any of them can carry: **a current or persistent claim additionally requires active currentness observation and evidence that re-evaluation occurred.**

Claim strength cannot exceed constitution independence.

An acceptance basis can be written so that it can never fail. Withdrawal requires a condition to change, and the producer usually writes the conditions. Define every condition as a fact about the moment of determination rather than a fact about the world, and nothing can ever go stale: invalidation latency stays at zero forever and the highest maturity level is satisfied vacuously. There is also no duty to look. Zero withdrawals and never checking produce identical reports.

Some facts are genuinely stable across a horizon, so the cure is not to manufacture an artificial failure condition. Currentness is an observed state. **A claim of current or persistent acceptance must identify the conditions whose change would invalidate reliance, the observation source responsible for detecting those changes, and the re-evaluation rule.** Where the authority determines that no material dynamic dependency exists for the stated horizon, that determination is itself explicit and reviewable.

Zero withdrawals is not evidence that nothing became stale. It is evidence of persistence only if the conditions capable of invalidating the measurement were observable and remained satisfied. Nothing changed and nobody looked are different measurements.

One gap the Six Locks do not close is benchmark optimization. A measurement can satisfy all six and still describe a system tuned to the evaluation set. Benchmark integrity is a separate control layer, not a missing seventh lock.

The Six Locks are the core measurement-boundary controls. They are not a complete benchmark-integrity program, and they are not a substitute for someone outside the measured party checking that the declarations were honest. A public benchmark claim additionally requires sealed workloads, work-mix control, contamination disclosure, pre-declared stopping rules, uncertainty treatment, held-out evaluation, and reporting of protocol deviations.

10. The Accepted-Work Measurement capability

No organization moves from counting tokens to governed accepted-work economics in a quarter, and none should try.

FIGURE 7

The Accepted-Work Measurement capability

Not a ladder every work class must climb.

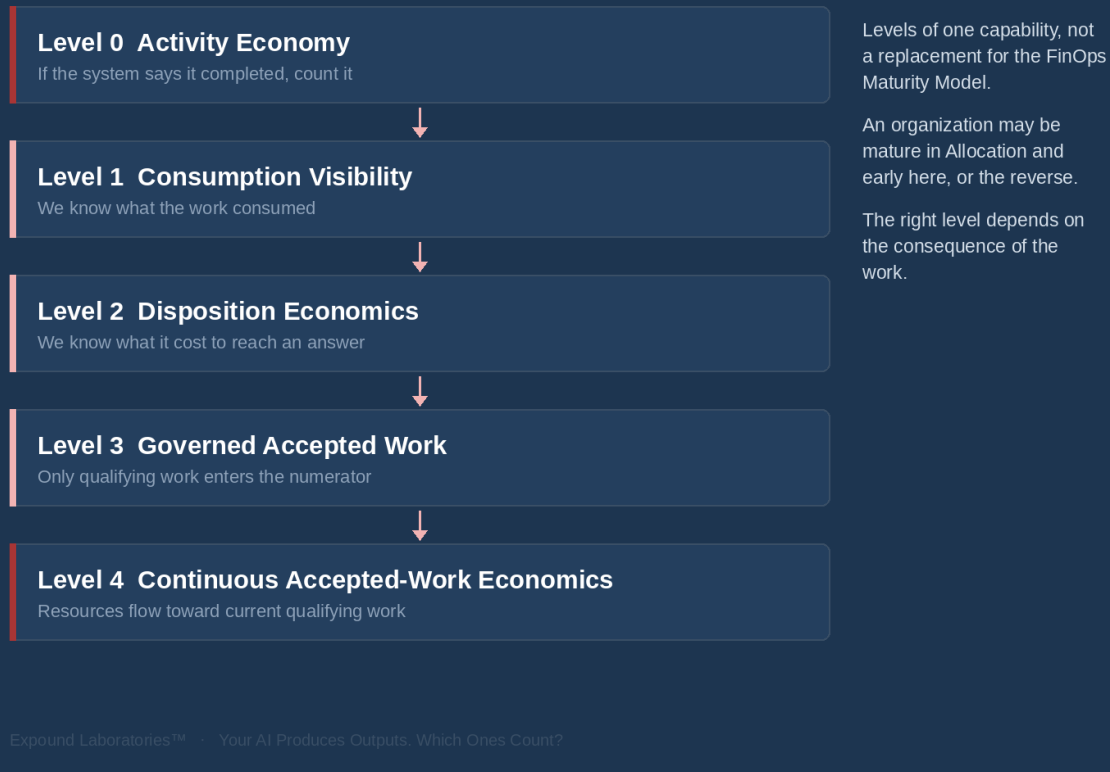


Figure 7 The Accepted-Work Measurement capability

What follows describes levels of a single capability, **Accepted-Work Measurement**. It is not a replacement for the **FinOps Maturity Model** and not a rival to it. The FinOps model is capability-specific and explicit that not every capability should be driven to maximum maturity. An organization may be mature in Allocation and early in Accepted-Work Measurement, or the reverse, and neither is a contradiction. The appropriate level depends on the consequence of the work measured.

The natural home for this is the **Intersecting Disciplines** capability, which coordinates FinOps with practices such as IT security, IT service management, enterprise architecture and platform engineering, sharing data, terminology and responsibilities without asking FinOps to absorb them. Acceptance authority usually sits with a domain owner, often with quality or audit. Those are not named in that capability, but the pattern holds: FinOps supplies the cost side, far better than this paper could specify, and the domain authority supplies what qualifies.

Level 0: Activity Economy. *"If the system says it completed, count it."* Spend is visible in invoices or broad cost centers. AI is measured by total bill or total tokens. There is no constituted work unit, no eligible population, no governed acceptance and no complete cost boundary. The first investment is not measurement but ownership and workload identity: you cannot measure a work class you cannot name.

Level 1: Consumption Visibility. *"We know what the work consumed."* Allocation, forecasting, model and token telemetry, workload classification, caching and capacity management are in place. Many organizations operating AI at scale will recognize themselves here. This paper does not claim a measured maturity distribution. Candidate work units exist, but completion still defines output. The next investment is attempt identity, eligible population and the capture of tool, human and failure costs.

Level 2: Disposition Economics. *"We know what it cost to reach an answer."* Fully loaded cost now includes models, infrastructure, tools, people, retries and shared services. Eligible, attempted, refused, escalated, accepted and indeterminate states become visible and reconcile. Acceptance may still be manual or self-authored. That is fine at this level and should be named as a limitation rather than papered over.

Level 3: Governed Accepted Work. *"Only qualifying work enters the numerator."* A governed acceptance mechanism establishes what may count, with authority appropriately separated from production. Measures: AWpD, cost per accepted work unit, accepted coverage, false acceptance, false refusal, evidence completeness, Time-to-Accepted-Work. Note that false *refusal* belongs here as prominently as false acceptance. A system that refuses too much is destroying value, and a framework that only measures one direction will drive exactly that failure.

Level 4: Continuous Accepted-Work Economics. *"Resources flow continuously toward current qualifying work, and no optimizer is permitted to redefine what qualifies."* Acceptance is recomputed as conditions change. Evidence reuse is scoped and current. Comparison is claim-specific. Portfolio economics operate on accepted work.

How the KPI stack expands across those levels. Nothing on the left becomes obsolete; the right is added beneath it.

FIGURE 8

How the KPI stack expands with accepted-work maturity

Upstream measures are retained as inputs. The accepted-work view is added beneath them, not substituted for them.

RETAIN	ADD
Token consumption and cost	Accepted Work and Cost-to-Disposition
Calls and tasks	Eligible and attempted work
Completed outputs	Typed dispositions
Cost per task	Cost-to-Disposition across every path
Producer success rate	Accepted Work under a separated authority
Raw throughput	Accepted coverage
Output per dollar	AWpD
Mean latency	Time-to-Accepted-Work
Human review burden	Accepted Work per expert hour
One aggregate score	AWpD beside coverage, integrity, time, risk, persistence and value

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Figure 8 How the KPI stack expands with accepted-work maturity

That last row is the one people skip, and the one that keeps the framework honest. **AWpD is a vector component, not a verdict.**

Each level has a characteristic way of going wrong:

FIGURE 9

Primary gaming risk by level

LEVEL	PRIMARY GAMING RISK
0 Activity Economy	Counting whatever the system emitted; activity read as productivity
1 Consumption Visibility	Upstream efficiency reported as productivity improvement: the claim-laundering case
2 Disposition Economics	Cost boundaries that quietly move; failures and retries dropped from the denominator
3 Governed Accepted Work	Acceptance loosened under delivery pressure; producer self-grading reintroduced as "pragmatism"
4 Continuous Accepted-Work Economics	Constitution or optimizer pressure redefines what qualifies; currentness is not re-evaluated

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Figure 9 Primary gaming risk by level

The levels are not a ladder every work class must climb. A work class where nothing consequential follows from the output should stay at Level 1 forever. A work class where a regulator, a customer or a payment depends on the result should reach Level 3 even if the volume is modest. **The right question is not "what level are we at" but "which work classes are at the wrong level for their consequence."**

The capabilities that carry an organization between levels fall into three groups. **Foundation:** work constitution, eligible population and coverage, consumption and cost visibility, disposition taxonomy. **Acceptance:** acceptance definition, acceptance authority, evidence and provenance captured during work rather than reconstructed afterward, and currentness. **Measurement and operating:** Cost-to-Disposition, comparability and claims, conformance and anti-gaming testing, service capacity, human attention, portfolio value, and federation.

The end state is **not** maximum automation, not zero human judgment, not maximum AWpD everywhere, and not governance over every draft and exploratory act. It is this:

Every consequential unit of work has an identifiable scope, an acceptance basis, evidence, a current disposition, an attributable cost, a relying-party context and a bounded claim. No optimizer is permitted to redefine what qualifies.

11. AI first, not AI only

The structure of the problem is older than AI, and noticing that turns a metric into a category.

Return to the principle from section 6: **the need for a reliance decision follows the consequence, not the identity of the executor**. The same discipline should apply wherever consequential work is produced, and be recognizable to practitioners in those domains, not an AI framework awkwardly exported.

The four domain sketches below are illustrative, not case studies. No organization's results are reported in them.

A software release. *Before:* CI is green, the release ticket closes, deployment frequency rises, audit evidence is reconstructed later, and a stale approval or changed dependency does not alter the release status. *After:* a release counts when the declared conditions are *current for the target environment*: evidence, authority, artifact identity, occurrence and no unresolved blockers. This is adjacent to existing work: DORA measures delivery throughput and instability, SLSA provides provenance and artifact-verification concepts, and neither is displaced. Accepted-work economics adds the *economic* question: what did it cost to reach a defensible release decision, and how much of what we shipped qualified.

An invoice approval. *Before:* workflow status says approved; throughput is invoices per employee; exceptions live in email. *After:* the invoice counts when supplier validity, purchase-order match where required, receipt or service evidence, pricing and tax checks, duplicate checking, current approval authority and payment eligibility are all established. No AI is involved here.

The same shape appears across the installed estate:

FIGURE 10

AI, non-AI and hybrid work classes

Illustrative work classes, not field results. The same discipline, applied wherever consequential work is produced.

WORK CLASS	COUNTED TODAY	ILLUSTRATIVE ACCEPTED WORK BASIS
Batch release	Batches produced; yield; cycle time	Batches meeting all predetermined release conditions under the applicable authority
Payment settlement	Transactions processed; straight-through rate	Correctly authorized, finally usable transactions under the governing rules
Healthcare prior authorization	Requests handled; approval rate	Valid determinations for a named patient, order, payer rule and current evidence state
Security finding closure	Findings closed	Findings whose remediation is evidenced and whose closure conditions still hold
Financial reconciliation	Items matched	Items matched with retained evidence sufficient to defend the match
OT control transition	Commands succeeded; uptime	Authorized, observed, outcome-supported transitions within the operating envelope

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Figure 10 AI, non-AI and hybrid work classes

The left column is what the operating report shows; the right is what someone downstream relies on, and the gap between them is where Cost-to-Disposition lives.

I am not claiming the non-AI opportunity is larger than the AI opportunity. I believe the installed estate is enormous, but I have no field evidence for a size comparison. That is a hypothesis requiring breadth and materiality studies, and it is registered as one.

What I am claiming is more modest:

AI supplies the urgency. The installed non-AI estate supplies the breadth. Hybrid work supplies the ordinary enterprise reality.

The AI-versus-non-AI framing is a false binary: AI drafts or recommends, deterministic systems retrieve and calculate and execute, humans resolve exceptions, workflow systems coordinate, and some named party relies on the result. **AWpD measures complete architectures, not models.**

12. Cross-domain understanding before cross-domain ranking

Section 15 states plainly that cross-domain normalization is experimental. That leaves a question:

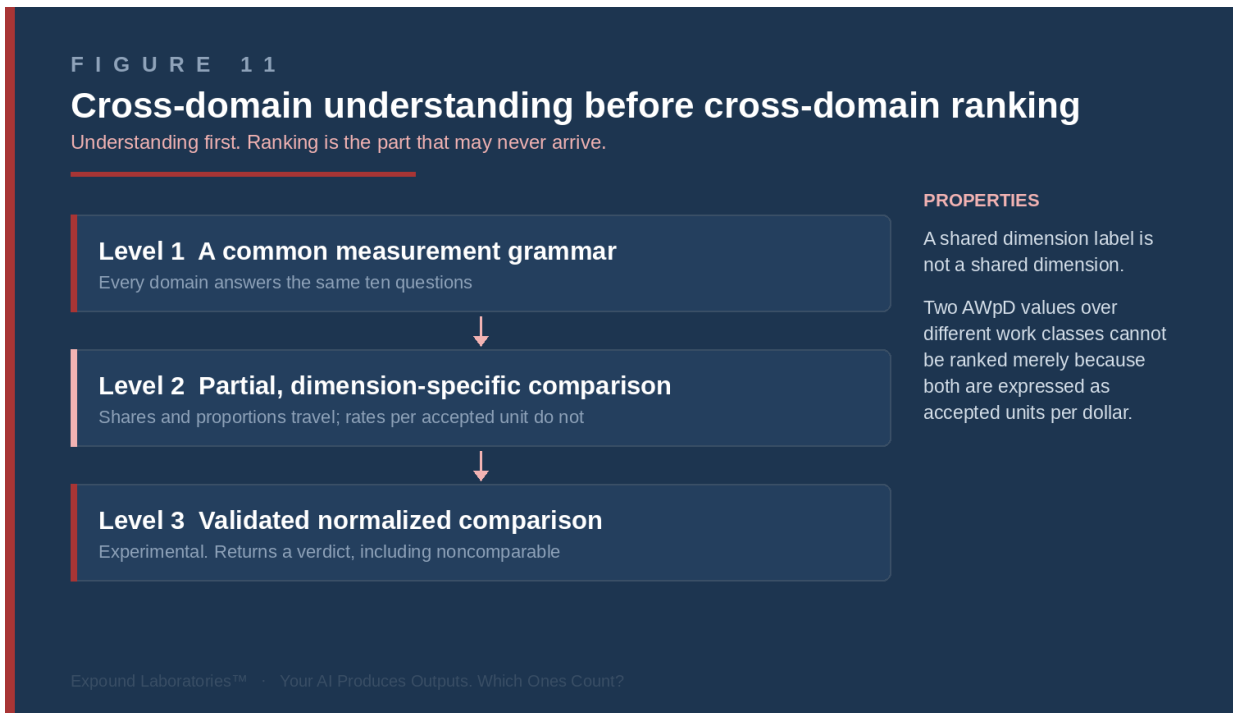


Figure 11 Cross-domain understanding before cross-domain ranking

What can accepted-work economics already do across domains, before any universal unit of work exists?

The answer rests on a distinction:

AWpD provides common semantics before it provides common units. Common grammar, local units.

Tier 1: a common measurement grammar

Available now, without any normalization research.

Every domain gets asked the same ten questions. What is one unit of work? What work was eligible? What was attempted? What makes it accepted? Who or what may establish acceptance? What evidence is required? What did reaching the disposition cost? What is the coverage? How long does acceptance remain current? What claim does the evidence permit?

An invoice, a software release and an AI recommendation keep entirely different domain meanings, but the organizations running them describe their economics and governance in the same categories, which makes each domain's *structure* legible to someone outside it.

Tier 2: partial and dimension-specific comparison

Some things compare even when total work does not.

Accepted coverage. Retry and failure cost, and verification burden, each as a share of disposition cost. The proportion of disposition cost spent reconstructing evidence not captured at the time. Time to disposition and time to Accepted Work. Invalidation latency. Acceptance-evidence maturity. Which of the Six Locks are absent. Direction and size of improvement against each domain's own baseline (an ordinal statement about that domain, not a cardinal one across domains).

Note what is *not* on that list. Human minutes **per accepted unit** is a within-work-class measure, not a cross-domain one: its denominator is the accepted unit, the object this section calls incommensurable across domains.

A shared dimension label is not a shared dimension. Two domains both reporting "verification burden" have a comparable dimension only where their definitions, cost boundaries, horizons, acceptance bases and evidence classes are compatible *for the claim being made*. Where they are not, PARTIAL, INCONCLUSIVE and NONCOMPARABLE are the correct verdicts at **this** tier too, not only at the experimental one.

Tier 3: validated normalized comparison

This is the experimental program for cross-domain normalized comparison (AWpD-X) and a universal cardinal work unit (ACWU), both defined in section 15. It requires validated work constructs, scale semantics, calibration, uncertainty treatment, invariance testing held-out-domain transfer, independent reproduction and evidence supporting the specific comparative claim.

Its correct outcome may be DIRECT, PARTIAL, COMPARABLE ONLY UNDER STATED NORMALIZATION, INCONCLUSIVE, NONCOMPARABLE, or a vector rather than a scalar. **Work may be irreducibly multidimensional, and failing to produce one number can be the scientifically correct result.**

What this looks like in a real enterprise

Two AWpD values computed over different work classes cannot be ranked against each other merely because both are expressed as accepted units per dollar. An accepted invoice is not an accepted software release. The shared metric name is not a shared unit.

But an enterprise may discover that both processes spend a large share of Cost-to-Disposition reconstructing evidence never captured at the time, let the producing workflow declare

completion, omit difficult unattempted work from coverage, and retain stale acceptance after dependencies change.

That does not prove an invoice and a release are equivalent work. **It identifies a common enterprise capability gap.**

Across domains whose definitions and cost boundaries are established as compatible for that question, the CFO can ask the question that justifies the whole exercise: **where would one shared evidence or conformance capability improve several domains at once?**

A portfolio view falls out without any universal unit:

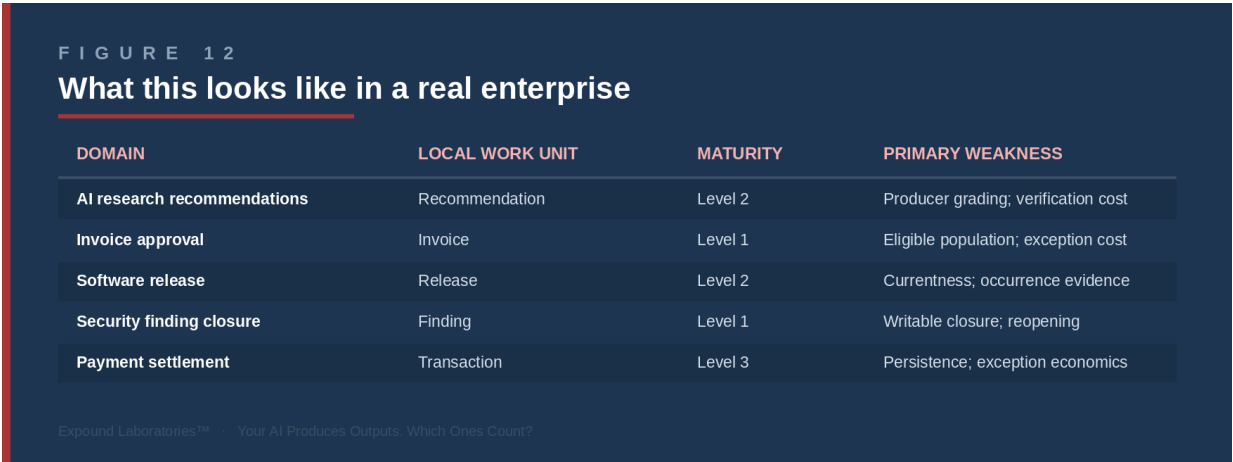


Figure 12 What this looks like in a real enterprise

No AWpD figures are compared in that table. It still tells a large enterprise where to invest, in what order, and which domains need a governed acceptance layer first.

The failure of universal AWpD-X would not invalidate AWpD-W. Within-work-class accepted-work economics stands on its own. Cross-domain normalization is a separate research question with a separate evidentiary burden, and the stable core does not depend on its success.

13. What this lets you manage

Measurement that changes no decision is overhead.

It exposes hidden cost. The first Cost-to-Disposition exercise routinely surfaces material costs paid outside the existing unit-economic view. How much is an empirical question.

It retains the economics of failure. Declined and failed work stops being invisible: a system that fails cheaply and fast is better than one that fails expensively and late.

It allocates premium capability where it earns its cost. Without an acceptance-qualified numerator, that argument runs on anecdote.

It reserves human judgment for judgment. Expert attention is the scarcest input in consequential work, and Accepted Work per expert hour makes it an allocated input rather than a residual. This is an *augmentation* measure: calling it labor replacement is analytically wrong, because the counterfactual is frequently no work at all. It is also a reliable way to lose the cooperation you need.

It governs delegation and portfolio choice. Work classes that could not previously be delegated become testable: define acceptance, run in shadow, measure accepted coverage, and expand where it holds. Accepted Work per budget and marginal AWpD then show which work classes deserve the next increment of spend.

The channels through which this work creates value overlap, and must not be summed. Every such total I have seen double-counts. AWpD measures one, throughput of qualifying work, and is silent on the rest.

What large enterprises will have to solve

Contract economics. Real cost bases are not list prices. Commitments, credits, reserved capacity, negotiated rates and shared services produce different numbers, so more than one view is needed: marginal, fully loaded and invoice-reconciled for anything finance will sign. FOCUS-style cost data and TBM-style cost-pool attribution are existing machinery to consume rather than reinvent.

Service and capacity. A determination that arrives after it was needed is not accepted work. Queuing, deadlines, model quotas, GPU capacity and, most bindingly, expert reviewer availability limit how much qualifying work a system can produce. Time-to-Accepted-Work and deadline-qualified Accepted Work belong beside the ratio.

Currentness and persistence. Persistent AWpD asks how much of what was accepted *continued* to qualify. Reusable evidence cuts both ways: it lowers marginal disposition cost at scale and widens the blast radius when it proves defective. Reuse must stay scoped, fresh and under current authority; invalidation latency is an operating metric.

Federation, privacy and audit. Large enterprises cannot centralize the meaning of "done": domains have different acceptance conditions. The workable pattern is local domain authority over acceptance semantics with central interoperability of measurement semantics: identities, cost and evidence classes, comparability verdicts and claim boundaries. Add confidential pricing,

sensitive work records and a correction and supersession process for measurements that turn out to be wrong.

Sampling appears to contradict section 9, which requires that unevaluated work stay distinguishable from accepted work. Enterprise practice at scale requires risk-based sampling. Both are right, and reconcile only if the result types stay separate.

Sampling can estimate an aggregate accepted-work rate. It cannot convert an unevaluated item into individually Accepted Work.

Three results follow, and conflating them is the failure:

- **Counted AWpD.** Every item in the numerator individually satisfied the declared acceptance basis.
- **Estimated AWpD.** The aggregate accepted-work rate was estimated through a declared sampling design, and the figure carries its sampling basis and uncertainty.
- **Operational reliance.** Item-specific, and never inferable from an aggregate. A sampled program can tell a board how the portfolio is performing, not an operator that this particular output may be acted upon.

14. How to begin

None of this requires buying anything. A team can begin the diagnostic with a spreadsheet, one bounded work class and a time-boxed measurement period. The diagnostic is where most of the value is.

FIGURE 13

How to begin

One work class. Not a program.

Pick one consequential work class

Something where being wrong actually costs



Constitute the unit

Declare what one item is, before outcomes are known



Declare the eligible population

Eligible, attempted, unattempted, refused, indeterminate



Write the acceptance basis down

Predetermined, and separate from the producer



Draw the cost boundary and horizon

Every attempt, every disposition, declared in advance



Report what the evidence permits

Including noncomparable, and including unknown

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Figure 13 How to begin

11. **Choose one recurring, consequential work class.** Not the whole enterprise. One class where something downstream acts on the result.
12. **Define one unit of work.** Write down what counts as one, including how partial and decomposed work is handled.
13. **Identify the eligible population.** All the work that *should* have been attempted, independent of what the system was willing or able to attempt.
14. **State what must be true before it counts.** In advance. In writing. Before you look at results.
15. **Identify who or what may establish that.** Name the authority, and note honestly where the producer is grading itself.
16. **Measure the complete Cost-to-Disposition.** Include failures, retries, verification and human time. Expect this step to be uncomfortable.
17. **Count only qualifying work.** Report the unqualified remainder rather than discarding it.

18. **Report coverage, integrity, service and limitations beside the ratio.** Never the ratio alone.
19. **Try to game your own measurement.** Take an afternoon and attempt every attack in section 9 against your own numbers.
20. **Use what breaks to build the roadmap.** The locks that failed are your capability backlog, in priority order.

Step 9 is the one to insist on. An organization that cannot game its own measurement in an afternoon has not understood it yet; one that games it easily has learned something far more valuable than a number.

15. What AWpD does not establish

A measure is only useful if you know what it does not measure. These are boundaries on the reading, not doubts about the instrument.

AWpD does not establish greater business value. It measures throughput of qualifying work, not what that work was worth.

It does not establish safety. A high ratio is compatible with a dangerous failure mode at low frequency.

It does not establish coverage. And **high AWpD with narrow coverage may be worse for the business than lower AWpD with broad coverage.**

It does not establish causation. Ratios move for many reasons, including work mix.

It does not establish generalization. A benchmark result does not transfer to production without rebasing, and a result in one work class says little about another.

It does not establish persistence. Point-in-time acceptance can overstate retained work where acceptance-relevant conditions change.

It does not establish the correct management objective. Efficiency is one input to a decision that also involves risk, service, capability and strategy.

And two scope boundaries within the framework itself. **AWpD-W** (within a single, declared work class and measurement basis) is the stable core, and it is what this paper is about. **AWpD-X**, cross-domain normalized comparison, and **ACWU**, a universal cardinal work unit, are *experimental research directions*. I am not proposing universal cross-domain leaderboards, and I would treat any such leaderboard built on this framework as a misuse of it.

And one non-claim about the ratio itself, which matters more than it may seem.

The arithmetic act of dividing an outcome by cost is not the contribution asserted in this article.

The differentiation asserted here is the architecture governing what enters the numerator, whether acceptance remains current, which costs remain attributable, and what comparisons the evidence supports.

I draw that boundary deliberately. An overbroad claim is the easiest thing in the world for a critic to defeat, and defeating it would leave the actual argument untouched while discrediting the paper that carried it.

16. Why computed reliance changes the economics

AWpD is not an output divided by cost. Its defining term is Accepted Work, and Accepted Work depends on a separate computed decision.

A result may be properly generated, transmitted, stored or executed without establishing that a particular party may rely on it for a particular downstream decision. Those are different questions. Finality Assurance asks the second: what may this named party rely upon, about this named result, for this named purpose, under the conditions that apply now?

That question can arise after execution. The result may already exist, and may already have been transmitted, filed or acted upon. No pre-execution gate, protected hardware, or special execution machinery is required for the reliance question to be answerable. Most organizations will not rebuild their execution path just to measure it.

Reliance is also not global. The same result may be sufficient for one party or purpose and insufficient for another. A summary may qualify for internal planning and not for a regulated filing. A recommendation may be suitable for review and not for direct action. The result has not changed. The reliance question has. A measurement that treats acceptance as a property of the output, rather than a relation between output, party and purpose, will overstate what it has.

Nor is reliance permanent. The original decision remains in the record. But if the evidence, authority, policy, dependency or permitted purpose changes, present reliance may narrow or be withdrawn, and the original act need not be reversed for current reliance to end. Historical validity and current applicability are separate, and most systems conflate them because nothing forced them apart.

Only work the current reliance decision permits enters Accepted Work. If that basis later fails, the Accepted Work record and its economics must be restated. That is uncomfortable, and it is the point. A measure that cannot be restated downward is not measuring what it claims. Every failed,

refused, escalated, abandoned, retried, verified, corrected and remediated path stays attributable in Cost-to-Disposition.

That complete loop is the distinction. Finality Assurance computes current reliance. Accepted Work admits only work that reliance permits. Cost-to-Disposition preserves every attributable route to resolution. AWpD measures the economic yield. Take away any one and what remains is a productivity ratio with good intentions.

I have looked for a system that closes this same loop and have not found one. Section 17 states that finding, its evidentiary basis and its bounds.

This section is conceptual. It does not disclose the reserved implementation.

17. Where this goes

The **Finality Assurance Family of Standards** is available for licensing under commercial terms. It is the acceptance architecture described in section 6: the computation of a present, scoped, evidence-backed reliance decision for consequential work, for organizations that need a reliance layer they do not have to build. Implementation, integration, domain-profile, conformance and design-partner pathways are available.

For AI and hybrid execution, **GMRS with Constrained Policy Reinforcement Learning** is a governed routing system in which learned preference cannot weaken permission, and in which AWpD operates as an economic signal among *admissible* routes. It has been through multiple prototype generations and is being finalized for product release. **A public open-source release is planned, with final license terms subject to legal and release review.** Treat the licensing and timing as forward-looking rather than committed. Non-AI adoption of accepted-work economics does not depend on it.

On rights, as precisely as I can make it.

Based on the public patent, academic, standards, vendor and commercial materials I reviewed through August 2026, I did not identify another publicly disclosed system that computes the same current, party- and purpose-scoped reliance decision or closes the same reliance-to-AWpD economic loop. I believe the architecture is materially differentiated. Undiscovered work cannot be ruled out, and this is not a legal conclusion.

The differentiation attaches to the whole composition: computed reliance governs numerator admission, acceptance can be withdrawn when its basis fails, every attributable path to a disposition stays in the denominator, and comparison is itself governed.

The architecture is the subject of pending patent applications and also includes separately retained trade secrets, copyrighted materials and marks. I intend to protect and enforce applicable rights. The objective is adoption through authorized pathways, not the isolation of ideas from legitimate use.

On publication and rights. I would rather be corrected in public than quietly wrong for another five years.

The concepts, questions, examples and diagnostic vocabulary in this article are published for discussion, testing, teaching and critique. Nothing in this publication grants a license under any patent or patent application, copyright, trademark, or other intellectual-property right. Certain implementation mechanisms remain confidential and are not disclosed here. Evaluation, implementation and commercial licensing pathways are available separately. Any future open-source release will be governed by the license accompanying that release.

None of which is meant to close the discussion.

That is the commercial position, and it is last for a reason. If the argument about measurement is wrong, none of it matters.

Closing

FinOps taught a generation of practitioners to ask what the technology cost. Tokenomics is extending that discipline across AI's production, consumption and value chain, including labor and other costs outside the model invoice. Both are necessary, both are further along than the caricature suggests, and neither is what this paper is arguing against.

The question that remains open sits between them and the business result:

In the Tokenomics Foundation's own words, Tokenomics connects the total cost of AI, including labor, to its outputs. Accepted Work per Dollar connects that cost to the work the organization was entitled to count.

Cost per token is not wrong. It answers an upstream question. The output is not wrong either. It simply does not count yet. What counts is the work that survived the conditions set before anyone knew the answer.

And across a large enterprise, the first thing this discipline delivers is not a universal number. It is a shared way of asking. **AWpD provides common semantics before it provides common units**, which is why an invoice team and a release team can learn something from each other long before anyone can say what one release is worth in invoices.

So here is the test I would apply, and the one I would like to see applied to this framework:

Take one metric your organization trusts. Try to improve it without improving the underlying economics. Every path that works identifies a boundary the metric does not govern.

If you find such a path in AWpD, I would like to hear about it. That is not a rhetorical flourish. A measurement framework for consequential work earns its standing by surviving people who try to break it, and the ones who try hardest are the ones it will eventually protect.

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Terminology. **Cost-to-Disposition** and **Accepted Work per Dollar** appear in the author's prior published work and are not coinages of this article. The **Six Locks** are a teaching synthesis introduced here.

Barcelona. The account of the October 2018 workshop is first-person. Contemporaneous photographs of the boards are retained but are not published with this article.



About Expound Laboratories™

Meaning, made computable. Finality, made governable.

Expound Laboratories™ develops research, standards, and computational systems concerned with Semantic Computation™ and Finality Assurance™. The work is motivated by a distinction that becomes more consequential as computational systems become more capable: producing an output is not the same as establishing what that output means, and neither is the same as determining whether it has earned a defined status for reliance. Three questions that ordinary workflows collapse into one:

What was produced? ≠ What does it establish? ≠ Under what conditions may it be relied upon?

Semantic Computation concerns the explicit representation and computation of meaning-bearing structures: claims, evidence, context, requirements, obligations, dependencies, authority, policy, provenance, contradictions, permitted transitions. Finality Assurance addresses the separate question of when work has satisfied the applicable evidence, policy, authority, scope and temporal conditions for a governed disposition. The first develops the semantic state; the second governs its disposition. A refusal is a verdict, and issuing one is not the same as satisfying it.

Finality Verdict issued ≠ Finality satisfied

The mark is the natural exponential, e^x , read two ways. Mathematically, e is Euler's number and the function has the property that its rate of change equals its current value, a compact object implying a much richer structure of direct and indirect relationships, which is the representational analogy intended. Semantically, the e stands for Expound and the x for the unresolved subject: a question, a claim, a decision, a requirement, a system state. Read as a progression, the mark says: expound the unknown, compute the meaning, govern the finality. Where it is rendered with an uppercase or stylized E , that is a brand form of the lowercase mathematical e .

The analogy is representational and is not a claim about implementation. Nothing in this article computes a scalar or matrix exponential as its governing algorithm, and the mark should not be read as asserting that it does. The exponential also carries the limitation the architecture exists to

address: an exponential has no notion of completion, sufficiency, authority or reliance, and a system's ability to keep generating does not establish that its output is finished.

Expansion ≠ Finality Scale ≠ Correctness Confidence ≠ Authority

Expound means to unfold and make meaning explicit; Laboratories adds the requirement to test it: hypotheses, controlled computation, evidence and provenance, measurement, adversarial testing, reproducibility, falsification, refusal. The combination is deliberate in both directions. Semantic elaboration without experimental discipline amplifies a wrong premise, and a closure mechanism operating over underspecified semantics authorizes a result whose meaning was never established. Finality here is not permanent or context-free certainty; it is an earned status inside an explicitly bounded jurisdiction of reliance, which is why every result is reported with the estate that produced it.

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FinOps, Tokenomics, and Accepted Work per Dollar

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